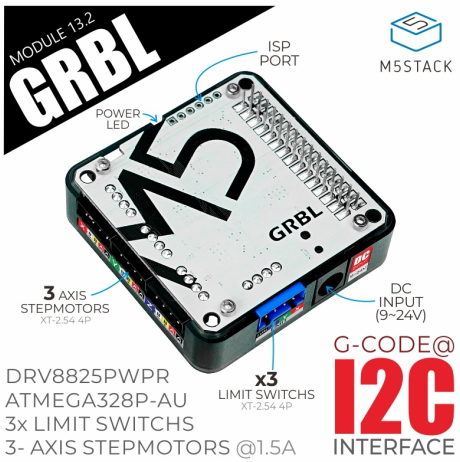
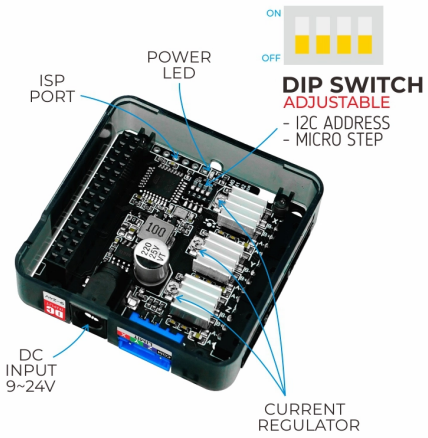
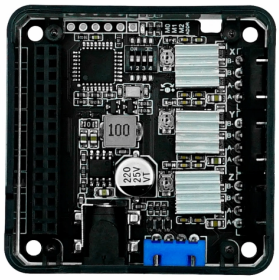
Module13.2 GRBL

SKU:M035



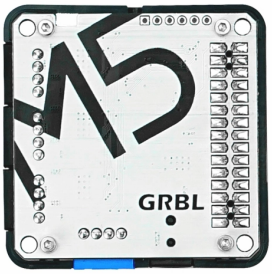












Description

Module13.2 GRBL is a three-axis stepper motor driver module in the M5Stack stackable module series. It uses the ATmega328P-AU controller paired with three DRV8825PWPR stepper motor driver chips, enabling simultaneous control of three bipolar stepper motors. It provides an I2C communication interface (addr: 0x70) and integrates a DIP switch for adjusting motor step subdivision (up to 1/32 microstepping) and I2C address configuration (supports dual addresses 0x70 and 0x71). This means you can achieve six-axis control by

stacking two **Module13.2 GRBL** modules. The power input interface supports DC 9~24V, with a motor drive current of up to 1.5A. It also features three sets of limit switch signal interfaces for external limit switches to enable motor braking (active low). It is suitable for various stepper motor motion control scenarios, such as printers, robotic arms, etc.

Features

- ATmega328P-AU controller
- Three-axis DRV8825PWPR stepper motor driver
- Drive current up to 1.5A
- Suitable for bipolar stepper motors
- Maximum 1/32 microstepping
- Limit switch interface (active low)

Includes

- 1 x Module13.2 GRBL

Applications

- Printers
- Scanners
- Office automation machines
- Factory automation
- Robotics

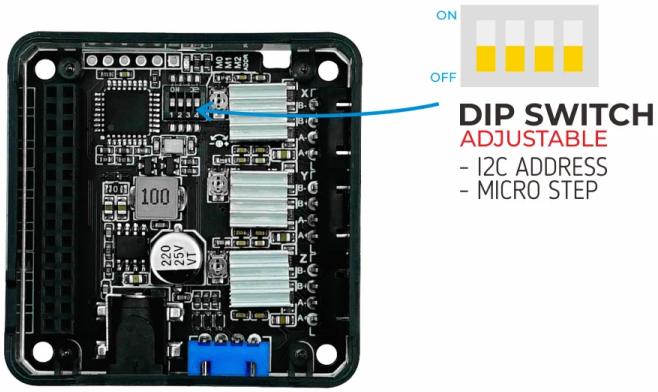
Specifications

Specification	Parameter
Motor Driver Chip	DRV8825PWPR
Controller Chip	ATmega328P-AU
Communication Interface	I2C Communication @0x70, 0x71
Max Driving Current per Channel	1.5A
Maximum Supported Microstepping	1/32
Interface Type	XT2.54-4P
Product Size	54.2 x 54.2 x 13.2mm
Product Weight	22.5g
Package Size	95.0 x 65.0 x 25.0mm
Gross Weight	42.3g

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Microstepping Configuration

MODE2	MODE1	MODE0	STEP MODE
0	0	0	Full step (2-phase excitation) with 71% current
0	0	1	1/2 step (1-2 phase excitation)
0	1	0	1/4 step (W1-2 phase excitation)
0	1	1	1/8 step
1	0	0	1/16 step
1	0	1	1/32 step
1	1	0	1/32 step
1	1	1	1/32 step

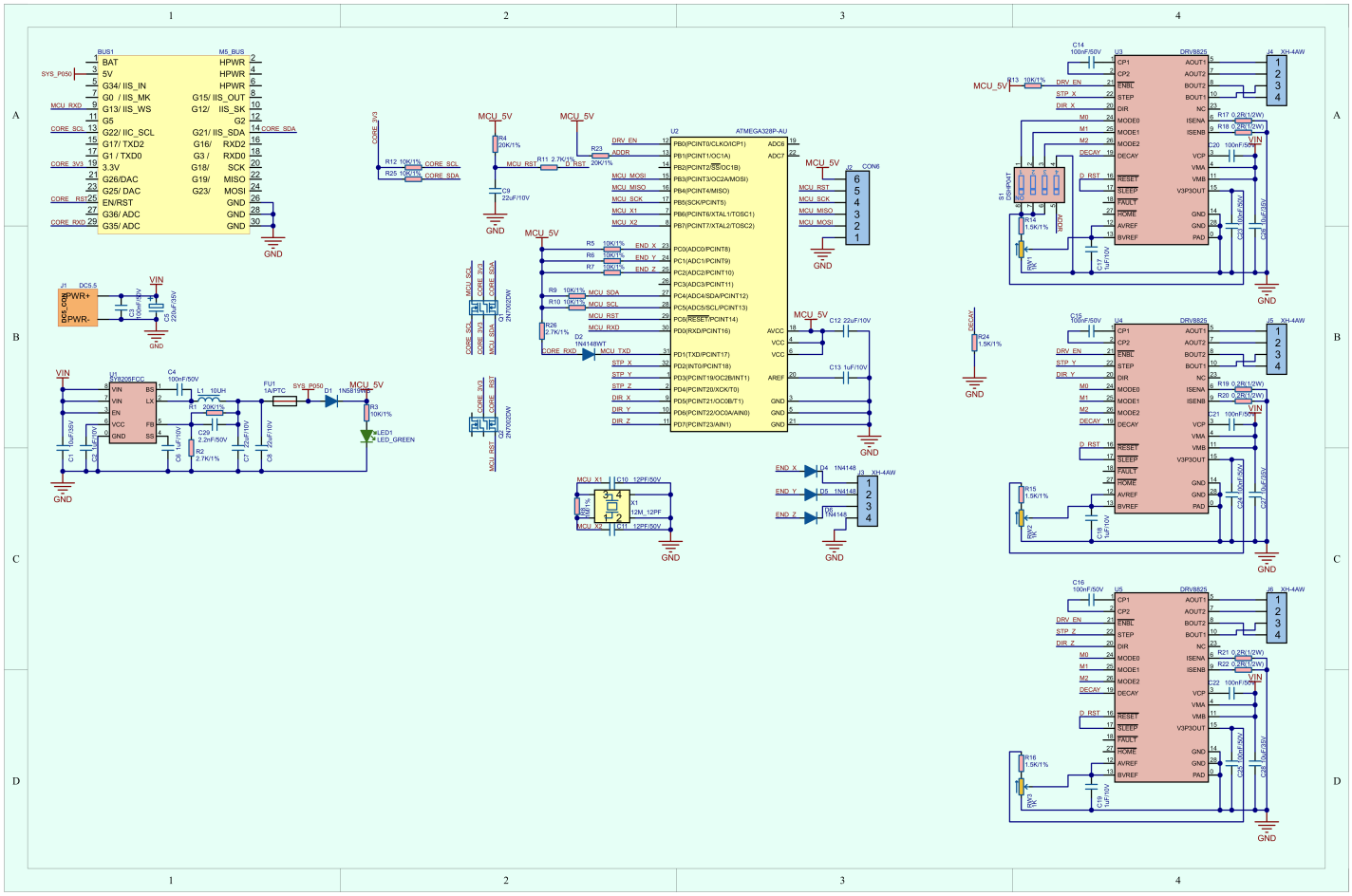


I2C Address Configuration

Switch	Address
0	0x70
1	0x71

Schematics

- Module13.2 GRBL Schematics PDF



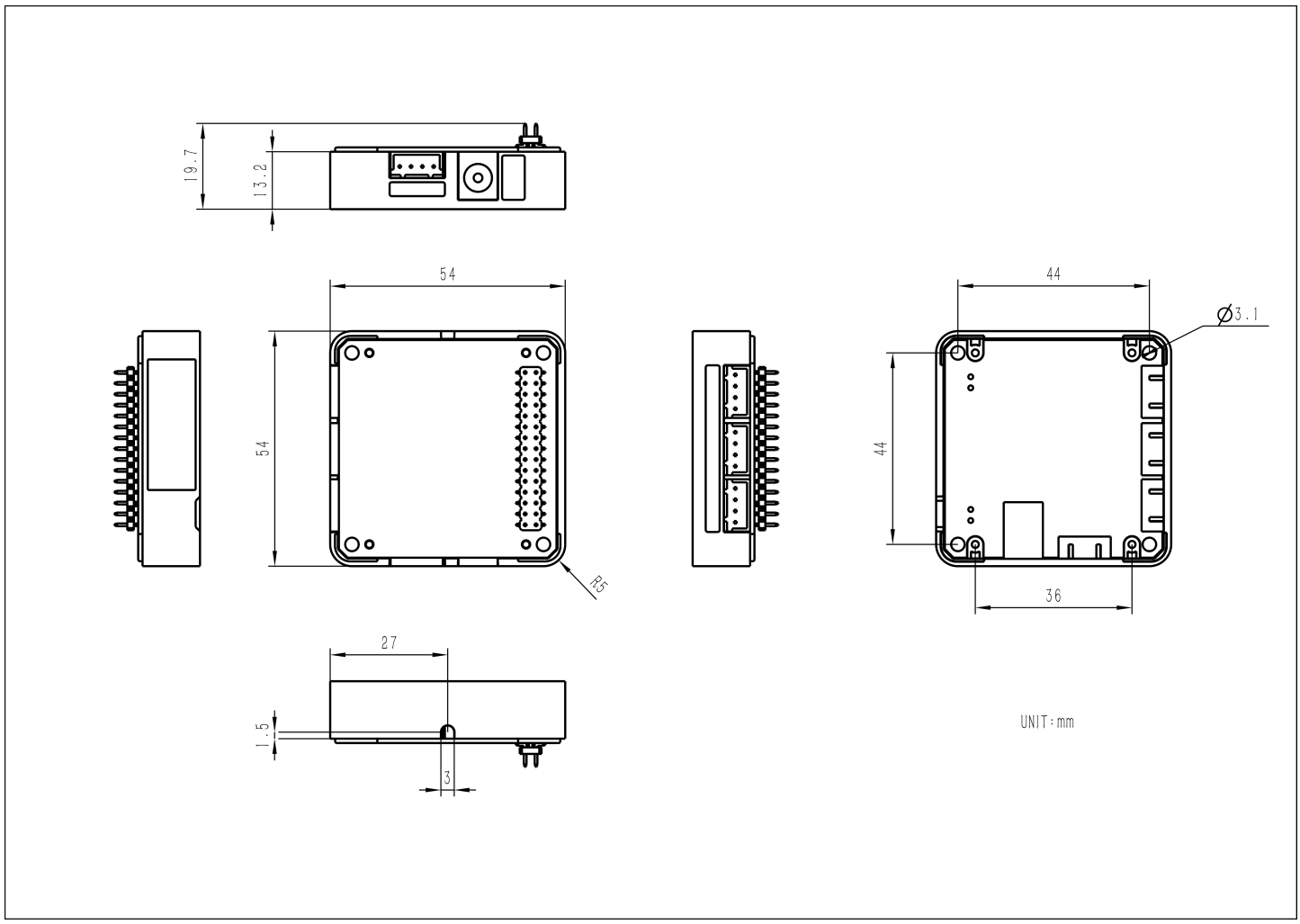
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	TXD
GND	3	4	
GND	5	6	RST
	7	8	
	9	10	
	11	12	3V3
	13	14	
	15	16	
SDA	17	18	SCL
	19	20	
	21	22	RXD
	23	24	
	25	26	
	27	28	5V
	29	30	

Model Size

- Module13.2 GRBL Model Size PDF



Datasheets

- DRV8825 Datasheet
- GRBL-Firmware
- GRBL v0.8 Configuration Reference Document

Softwares

Arduino

- Module13.2 GRBL Arduino Library

UiFlow1

- Module13.2 GRBL UiFlow1 Documentation

UiFlow2

- Module13.2 GRBL UiFlow2 Documentation

Easyloader

Easyloader	Download Link	Notes
Module13.2 GRBL Example Easyloader with M5Core	download	/

Video

GRBL13.2.mp4